

All local one-norm maximizers on $O(3)$

Candidate full solution to a concrete problem in arXiv:0901.2923

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Status: candidate full solution, likely valid. For the entrywise norm

$$\Phi(U) = \sum_{i,j=1}^3 |U_{ij}|, \quad U \in O(3),$$

there are exactly 192 local maximizers. They form one class under signed row and column permutations, represented by

$$U_* = \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{pmatrix}.$$

Every local maximizer is therefore global and has $\Phi(U) = 5$.

1 The source problem

Banica, Collins, and Schlenker prove that, if $U \in O(N)$ has sign matrix $S = (\text{sgn } U_{ij})$, then

$$U \text{ locally maximizes } \Phi \iff SU^T > 0. \quad (1)$$

The condition includes the fact, proved just before their criterion, that a local maximizer has no zero entry. After discussing tensor products, they state the concrete unresolved task of computing the local maximizers at $N = 3$, reducing it to two sign patterns over $SO(3)$, but report that the Euler–Rodrigues calculation gives no further result [1, p. 8]. The complete source page is reproduced below.

matrix $Y_{ia,jb} = S_{ij}X_{ab}$, and the result follows from the following computation:

$$\begin{aligned} (YW^t)_{ia,jb} &= \sum_{kc} Y_{ia,kc} W_{jb,kc} \\ &= \sum_{kc} S_{ik} X_{ac} U_{jk} V_{bc} \\ &= \sum_k S_{ik} U_{jk} \sum_c X_{ac} V_{bc} \\ &= (SU^t)_{ij} (XV^t)_{ab} \end{aligned}$$

(2) By using the equivalence relation, we can assume that the matrix is a rotation of angle $t \in [0, \pi/2]$. With $a = \cos t$, $b = \sin t$ we have:

$$SU^t = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} a+b & a-b \\ b-a & a+b \end{pmatrix}$$

Thus we have $a = b$, so $U = S/\sqrt{2}$, and S is Hadamard.

(3) Consider indeed the matrix $U = A \otimes (H/2)$, where A is the 3×3 matrix found in section 2, and H is a 4×4 Hadamard matrix. Then U locally maximizes the 1-norm on $O(12)$, but is not a multiple of an Hadamard matrix. $\square$

The local maximizers don't seem to be stable under more general operations.

One problem is to compute these local maximizers at $N = 3$. Observe that, by using the equivalence relation, we can assume that $U \in O(3)$ has the following sign matrix:

$$S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}$$

Equivalently, we can assume that $U \in SO(3)$ has the following sign matrix:

$$S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & \pm 1 \\ 1 & 1 & -1 \end{pmatrix}$$

So, in principle we have 2 concrete problems over $SO(3)$ to be solved. However, the computation with the Euler-Rodrigues formula seems to be quite difficult, and we do not have further results in this direction.

4. NORM ESTIMATES

Our purpose here is to investigate the following quantity:

$$K_N = \sup_{U \in O(N)} \|U\|_1$$

2 Proof intuition

There is no need to solve nonlinear Euler–Rodrigues inequalities. On any fixed orthant, the one-norm is the linear functional $\langle S, U \rangle$. Criterion (1) says exactly that a local maximizer is the orthogonal polar factor of its sign matrix. Thus the continuous problem becomes a finite sign problem.

For 3×3 matrices the finite problem collapses completely. Every nonsingular sign matrix is equivalent to one canonical matrix. Its polar factor has the required signs and passes the positivity test. Equivariance then proves that every nonsingular sign pattern works. Counting those patterns gives 192, and the canonical polar factor is the known global optimizer.

3 The polar codebook

Lemma 1 (polar reduction). *Let $S \in M_N(\{\pm 1\})$, and, when S is nonsingular, put*

$$Q(S) = (SS^T)^{-1/2}S.$$

Then $U \in O(N)$ is a local maximizer of the entrywise one-norm if and only if

$$S = \text{sgn}(U), \quad S \text{ is nonsingular}, \quad U = Q(S).$$

Equivalently, the admissible sign matrices are precisely those satisfying $\text{sgn } Q(S) = S$.

Proof. Suppose first that U is a local maximizer and $S = \text{sgn}(U)$. By (1), $P = SU^T$ is positive definite. Hence

$$S = PU, \quad SS^T = P^2.$$

The positive square root is unique, so $P = (SS^T)^{1/2}$ and $U = P^{-1}S = Q(S)$. In particular S is nonsingular.

Conversely, suppose S is nonsingular and $\text{sgn } Q(S) = S$. Directly,

$$Q(S)Q(S)^T = I, \quad SQ(S)^T = (SS^T)^{1/2} > 0.$$

Criterion (1) now makes $Q(S)$ a local maximizer. $\square$

Corollary 2. *For every fixed N , the local maximizers form a finite set, naturally embedded in the set of nonsingular $N \times N$ sign matrices.*

Proof. Lemma 1 assigns at most one local maximizer to each of the finitely many sign matrices. $\square$

The construction is equivariant. If A, B are signed permutation matrices, functional calculus gives

$$Q(ASB) = A Q(S) B. \tag{2}$$

Consequently admissibility and the corresponding local maximizers are preserved by the source's equivalence relation.

4 The 3×3 sign orbit

Lemma 3. *All nonsingular 3×3 sign matrices are equivalent under signed row and column permutations.*

Proof. By multiplying rows and columns by signs, normalize the first row and first column to $+1$. The matrix then has the form

$$S(a, b, c, d) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & a & b \\ 1 & c & d \end{pmatrix}, \quad a, b, c, d \in \{\pm 1\}.$$

Associate the binary matrix

$$X = \frac{1}{2} \begin{pmatrix} 1-a & 1-b \\ 1-c & 1-d \end{pmatrix}.$$

Subtracting the first row from the other two gives

$$\det S = 4 \det X.$$

Thus S is nonsingular exactly when the 2×2 binary matrix X has determinant ± 1 . There are six such matrices: the two permutation matrices and the four matrices $\mathbf{1} - E_{ij}$. Permuting the last two rows and columns reduces these to the two representatives

$$X_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Their normalized sign matrices are

$$S_1 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Finally, cyclically permuting the columns of S_1 in the order $(2, 3, 1)$, then multiplying its second row by -1 , gives S_2 . Hence there is only one full equivalence class. $\square$

5 Classification and exact count

Theorem 4. *The entrywise one-norm on $O(3)$ has exactly 192 local maximizers. Every one is equivalent to*

$$U_* = \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{pmatrix}.$$

In particular, every local maximizer is a global maximizer and has one-norm 5.

Proof. Take the canonical sign matrix

$$S_* = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$U_* U_*^T = I, \quad \text{sgn}(U_*) = S_*,$$

and

$$S_* U_*^T = \frac{1}{3} \begin{pmatrix} 5 & 1 & 1 \\ 1 & 5 & -1 \\ 1 & -1 & 5 \end{pmatrix}.$$

The last matrix has eigenvalues $1, 2, 2$, hence is positive definite. It follows from Lemma 1 that $U_* = Q(S_*)$.

By Lemma 3, every nonsingular sign matrix is equivalent to S_* . Equivariance (2) therefore shows that every nonsingular 3×3 sign matrix is admissible and that all resulting local maximizers are equivalent to U_* .

It remains only to count. The normalization used in Lemma 3 has six nonsingular binary tails. Each normalized matrix has $2^{3+3-1} = 32$ preimages under independent row and column sign changes; the exponent is reduced by one because simultaneous global negation on both sides acts trivially. Hence there are

$$6 \cdot 32 = 192$$

nonsingular sign matrices. Distinct sign matrices yield distinct maximizers, so the count is exact.

Finally, $\Phi(U_*) = 5$. The source proves that 5 is the global maximum of Φ on $O(3)$ [1, Theorem 2.1]; equivalence preserves the norm. Thus every local maximizer is global. $\square$

6 Checks, upgrade attempts, and scope

Independent finite check. Exhaustive enumeration of all $2^9 = 512$ sign matrices found 192 nonsingular matrices, one signed-permutation orbit. Exact symbolic multiplication verified $U_* U_*^T = I$, $S_* U_*^T > 0$, and $\Phi(U_*) = 5$. The enumeration was used only as a check; the proof above is combinatorial and exact.

Deep upgrade. The first route sought only the two $\text{SO}(3)$ cases displayed by the source. The polar reduction upgrades this to all of $\text{O}(3)$, proves there is one equivalence class, and supplies the exact count 192. Corollary 2 also gives a finite polar-codebook description in every dimension. The same reduction does not directly solve the source’s p -norm problem for $p \neq 1$, because the gradient inside an orthant is then entry-dependent rather than a fixed sign matrix. Nor does the 3×3 orbit collapse persist in higher dimensions, where many sign-matrix orbits occur.

Novelty check. Bounded primary-literature searches through 11 August 2026 covered the source title, the phrases *local maximizers*, *one-norm*, and $\text{O}(3)$, and later work on almost Hadamard matrices. Banica, Nechita, and Życzkowski [2] repeat criterion (1) and identify the optimal/global $N = 3$ matrices, but the inspected text does not state the classification or count of all local maximizers. No primary source recording the theorem above was located. Novelty confidence is moderate because the search was bounded.

Human-review recommendation. Check the implication from positivity to the polar factor in Lemma 1, the two-type reduction of binary 2×2 matrices in Lemma 3, and the $6 \cdot 32$ count.

References

- [1] T. Banica, B. Collins, and J.-M. Schlenker, *On orthogonal matrices maximizing the 1-norm*, Indiana Univ. Math. J. 59 (2010), 839–856; [arXiv:0901.2923](#).
- [2] T. Banica, I. Nechita, and K. Życzkowski, *Almost Hadamard matrices: general theory and examples*, Open Syst. Inf. Dyn. 19 (2012), 1250024; [arXiv:1202.2025](#).